Biotechnology

24/08/25

Basic Concepts –

Biotechnology (breakdown)

Biology → Study of living beings / life

Technology → Application of science for human welfare

Characterstics of life –

Life is total sum of biochemical reaction occuring in body or cell.

Characterstics of living beings –

1. Reproduction → Creating new individuals
2. Cell structure / Cellular organization → Composed of one or more cell.
3. Life processes → Metabolism → Biochemical reaction in cell / Body
4. Adaptation → Long term genetic changes → Evolution

Metabolism — Types

Catabolism
Breakdown reaction
Matter will breakdown & energy is liberated
e.g Respiration

Anabolism
Synthesis reaction
Energy will be consumed & matter would form
e.g Photosynthesis

Virus
↓
Connecting link between Living and Non-living
↓ Reproduction
↓ No cell structure, No Metabolism

Types

On the basis of Genetic materials

i) DNA virus → Genetic material in DNA
e.g Pox virus (Small Pox, Chicken Pox, Monkey Pox) HPV (Human papilloma virus), Hepatitis-B
ii) RNA Virus → Genetic material is RNA
Retrovirus → old virus
e.g HIV, Polio, Dengue, measles, Mumps, Hanta, Nipah, Flu (H1N1, H5N1), Influenza, Covid, Chikungunia, Zika

On the basis of Infection Target

i) Animal virus → Animal cells
e.g Influenza, Herpes Simplex virus
ii) plant virus → plant cells
e.g Tobacco mosaic virus
iii) Bacteriophages → Bacterial cells
e.g Bacteriophage T4
iv) Mycophages → Fungi
e.g CHV1

Zoonotic virus → Transfer from animals to human body e.g Coronavirus, HIV, Rabies

* Body immunity → ~~Interfera~~ Interferon protein → Help to fight disease

Cell structure –

1. Cell is fundamental unit of life.

2. Living organism are made up of cells
3. Functions of body are outcome of cell activity
4. New cells come from pre-existing cells by cell division
5. cell has many cell-organelles inside to perform different tasks.

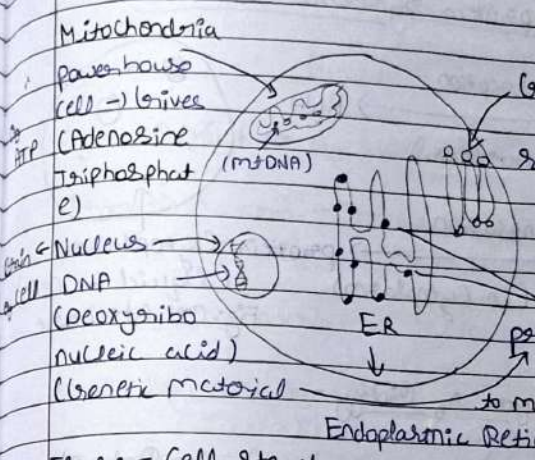


Figure – Cell structure with components

Protein → Building materials
Carbohydrates → Energy fuel
Fats and lipids → Stored fuel
Vitamins, Minerals → protective, save from disease

Protein Synthesis
↓
All our body, enzymes in our body and hormones are made up of proteins

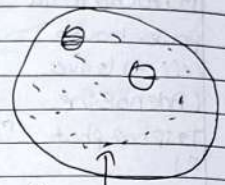
Process –

1. [DNA (Nucleus)] $\xrightarrow{\text{Information}}$ [RNA]
2. [RNA] $\xrightarrow{\text{pass the information}}$ [Ribosome]
3. [Ribosome] $\xrightarrow{\text{Make}}$ [protein]

Steps of protein synthesis in a cell -

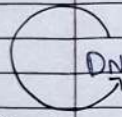
DNA $\xrightarrow{\text{Transcription}}$ RNA
(in nucleus)

RNA (with Ribosome) $\xrightarrow{\text{Translation}}$ protein
(in Cytoplasm)



Cytoplasm (liquid in cell)
Fig: Cell structure

Central Dogma of Biology -



DNA $\xrightarrow{\text{Transcription}}$ RNA $\xrightarrow{\text{Translation}}$ protein $\rightarrow$ Body structure
protein $\rightarrow$ [Body function by Enzymes and hormones] $\rightarrow$ Metabolism

Replication (during cell division)
Reverse Transcription (in Retro Virus)

Nucleus
↓
Controlles cellular activities, houses genetic information

Nuclear membrane
Nucleolus
Nucleoplasm (liquid)
Chromatin Material
↓ DNA — Genetic material
↓ Histone protein — Non-Genetic material

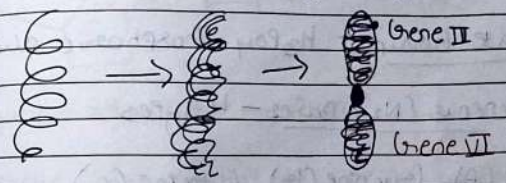
Figure: Structure of Nucleus

Chromatin material — Thread like structure present in nucleus

DNA - Genetic material in Nucleus

Gene - Segments of DNA

Chromosome - Coiled structure of DNA with histone (appear only during cell division)



↑ Coiling with Histone

Gene I Gene II Gene III Gene IV Gene V Gene VI

⟵ DNA ⟶

Genes → Unit of inheritance of Genetics

Polygenic effect – Skin colour, Blood group

DNA – Deoxyribonucleic Acid

Raw Material → Unit / Monomer → Polymer

Clay, sand & Water → Bricks → Wall

C, H, O → Glucose → Starch

{Sugar, phosphate Group, Nitrogeneous base} → Nucleotide → DNA

Raw Materials of DNA –

1. Sugar – Deoxyribose Sugar ($C_5H_{10}O_4$)

No. of Carbon is 5 → pentose Sugar

2. Phosphate Group – H_3PO_4 Phosphoric acid

3. Nitrogeneous (N_2) Bases – 4 Types

Adenine (A), Guanine (G), Cytosine (C), Thymine (T)

Unit of monomer of DNA –

↓

Nucleotide [C 4 Types – A, T, G, C] → Nitrogeneous base

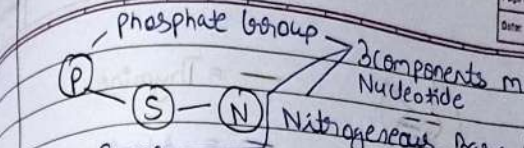


Figure – Nucleotide Structure and components of DNA

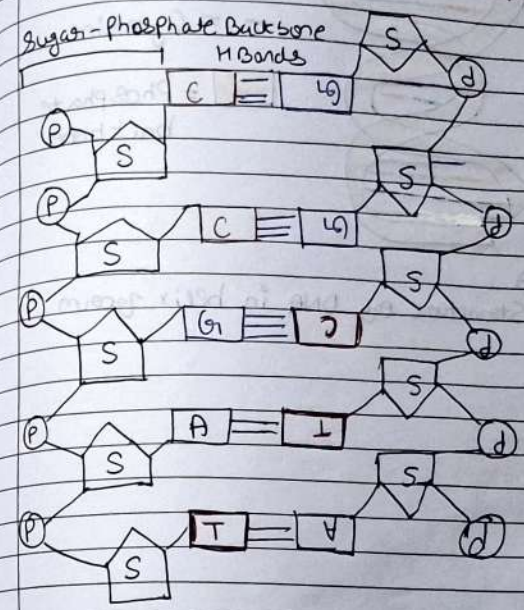
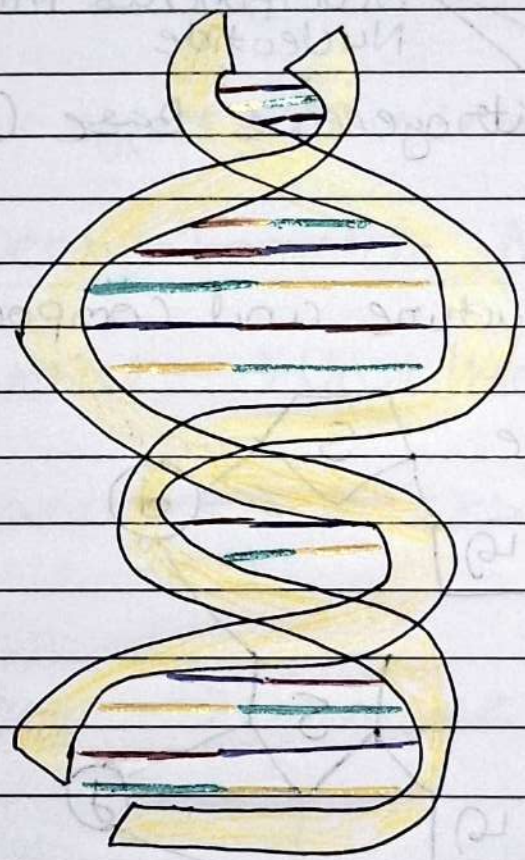


Figure – Structure of DNA

- Genetic information on DNA is written in the form of a sequence of Nitrogen base pairs
- Nucleotides are joined together by phosphodiester bonds, forming the sugar phosphate backbone that stabilises the DNA structure.



— = Thymine

— = Adenine

— = Guanine

— = Cytosine

▭ = Phosphate backbone

DNA

Figure – Structure of DNA in helix form